

Experiment No: 04

Machine Learning

Aim:

Train a Linear Regression model on Boston Housing Data, visualizing residuals vs predictions.

Objective:

- To understand linear regression modeling on real-world datasets.
- To train a regression model using the Boston housing dataset.
- To visualize residuals and predictions for evaluating model performance.

Prerequisites:

- Software: Python 3.x with libraries: numpy, pandas, matplotlib, seaborn, scikit-learn.
- Hardware: A computer with at least 4GB RAM.
- Skills: Basic Python programming and knowledge of regression models.
- Dataset Availability: Boston Housing dataset (CSV format).

Theory:

Linear Regression is one of the simplest supervised machine learning algorithms used for predicting a dependent variable based on one or more independent variables. The Boston Housing dataset contains information on housing prices and several features such as crime rate, number of rooms, and distance to employment centers. By fitting a linear regression model, we can estimate housing prices and evaluate model performance using residuals (the difference between actual and predicted values). Visualizing residuals against predictions helps check model assumptions and detect patterns indicating poor fit or non-linearity.

Apparatus / Software Requirements:

- Python IDE or Jupyter Notebook
- Boston Housing dataset (CSV)
- Internet connection for installing libraries (if required)

Procedure:

Step 1: Import required libraries (numpy, pandas, matplotlib, seaborn, scikit-learn).

Step 2: Load the Boston Housing dataset into a DataFrame.

Step 3: Explore the dataset and visualize feature-target relationships.

Step 4: Split the data into training and testing sets.

Step 5: Train a Linear Regression model using training data.

Step 6: Evaluate model predictions using Mean Absolute Error (MAE).

Step 7: Visualize Actual vs Predicted values and plot residuals vs predictions.

Program Code:

```
# Importing required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Importing scikit-learn modules
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error

# Load Boston Housing dataset
df = pd.read_csv("Boston_housing_data.csv")
df.head()

# Scatter plot: Number of Rooms (rm) vs House Price (medv)
plt.figure(figsize=(10,8))
sns.scatterplot(data=df, x="rm", y="medv", color="red")
plt.title("Number of Rooms vs Price of House")
plt.show()

# Selecting features and target
X = df.drop("medv", axis=1)
y = df["medv"]

# Splitting the data into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize and train the Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Predicting on test data
y_pred = model.predict(X_test)
```

```
# Mean Absolute Error
```

```
MAE = mean_absolute_error(y_pred, y_test)
```

```
print("Mean Absolute Error:", MAE)
```

```
# Plot Actual vs Predicted values
```

```
plt.scatter(y_test, y_pred, color="red")
```

```
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], color="black",  
linewidth=4, label="Perfect Prediction")
```

```
plt.title("Actual vs Predicted Values")
```

```
plt.xlabel("Actual")
```

```
plt.ylabel("Predicted")
```

```
plt.legend()
```

```
plt.grid(True)
```

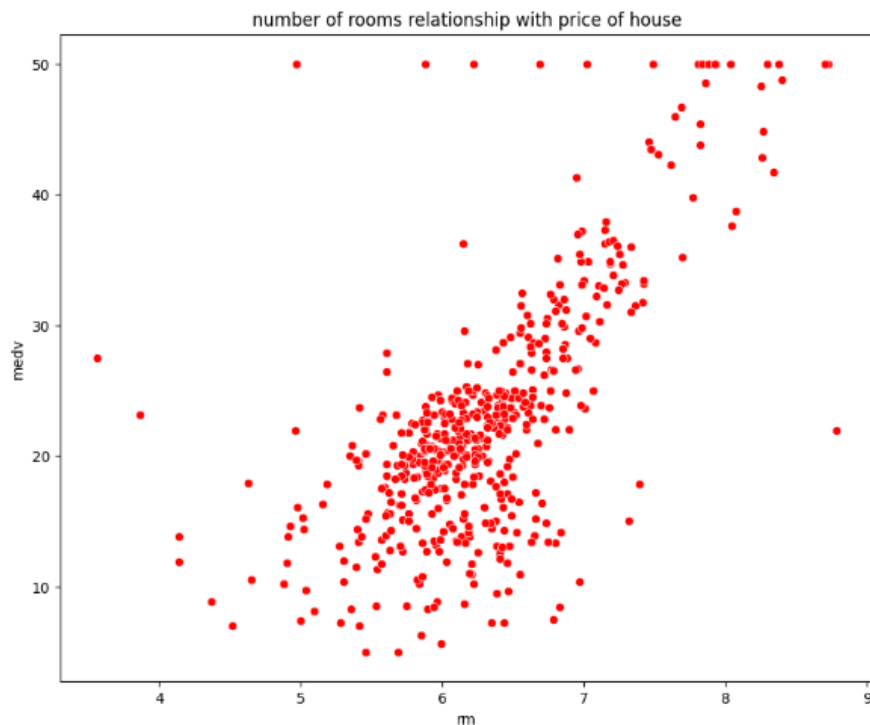
```
plt.show()
```

Expected Output:

1.

Unnamed: 0	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
0	1	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	24.0
1	2	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	21.6
2	3	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	34.7
3	4	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94
4	5	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33

2.



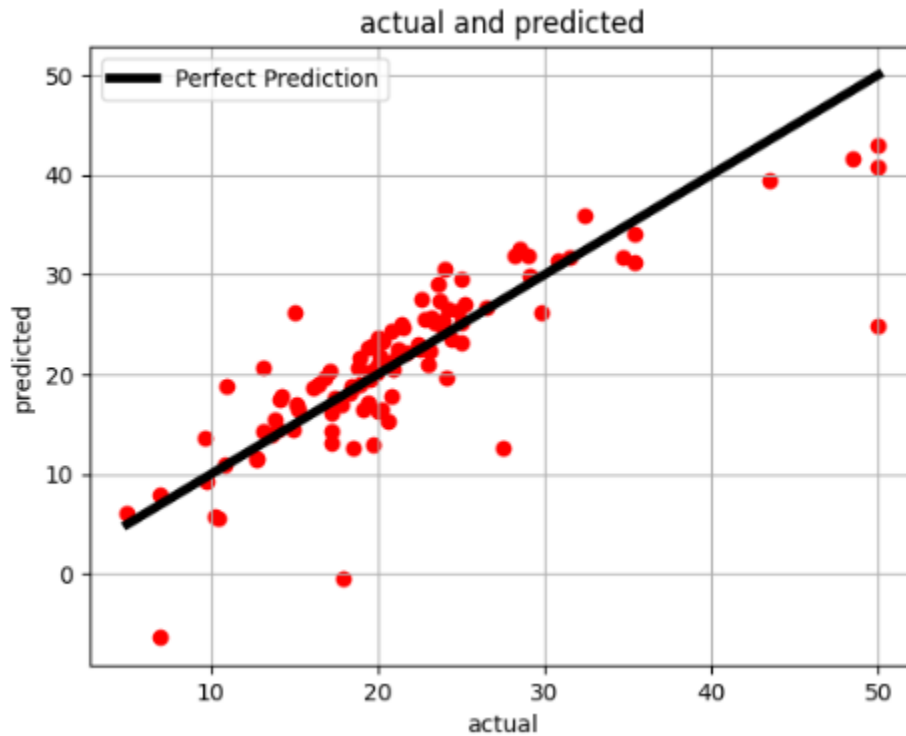
3.

	Unnamed: 0	crim	zn	indus	chas	nox	rm	age	dis	\
477	478	15.02340	0.0	18.10	0	0.6140	5.304	97.3	2.1007	
15	16	0.62739	0.0	8.14	0	0.5380	5.834	56.5	4.4986	
332	333	0.03466	35.0	6.06	0	0.4379	6.031	23.3	6.6407	
423	424	7.05042	0.0	18.10	0	0.6140	6.103	85.1	2.0218	
19	20	0.72580	0.0	8.14	0	0.5380	5.727	69.5	3.7965	
..	...	...	...	...	...	...	...	...	...	
106	107	0.17120	0.0	8.56	0	0.5200	5.836	91.9	2.2110	
270	271	0.29916	20.0	6.96	0	0.4640	5.856	42.1	4.4290	
348	349	0.01501	80.0	2.01	0	0.4350	6.635	29.7	8.3440	
435	436	11.16040	0.0	18.10	0	0.7400	6.629	94.6	2.1247	
102	103	0.22876	0.0	8.56	0	0.5200	6.405	85.4	2.7147	

4.

```
array([28.97335364, 35.87795794, 13.97366183, 25.61583978, 18.66608153,
       23.64466022, 17.05971239, 14.5264485 , 22.79402759, 19.60618783,
       24.62043501, 19.06428675, -6.27197735, 22.20788474, 18.64849181,
       26.15022718, 20.62835191,  5.77374284, 40.72117979, 17.48530737,
       27.08436097, 29.90818637, 11.65330429, 22.99879937, 17.85048447,
       15.5556251 , 23.31650588, 14.44013189, 22.11106256, 18.20109763,
       22.33391113, 24.79774986, 26.15304018, 17.89425941, 16.52927207,
       17.10592481, 31.77372627, 19.58243073, 23.603617 , 25.2202457 ,
       12.89702838, 31.91216634, 42.90231353, 17.74616574, 27.5000776 ,
       17.06259552, 14.39460992, 26.51167547, 20.10908837, 30.51587021,
       21.74636797, 34.09117218, 16.44234996, 26.62572161, 39.42027423,
       22.49131562, 18.75511558, 32.62323592, 25.35325167, 12.58127149,
       23.23090959, 31.15654212, 31.70938821, 16.555899 , 19.75136599,
       16.30198537, 20.60703158, 26.4253367 , 31.3483927 , 11.44469483,
       20.93853272, 27.33689972, 10.88471601, 15.24992839, 24.31201729,
       6.16749235, 21.87268675, 41.62910729, 18.88336507,  7.99630902,
       21.29335313, 13.20061512, 20.55490914,  9.36676224, 22.6651137 ,
       31.82868525, 18.98610512, 25.13454836, 29.56561095, 20.35948068,
       25.6740408 ,  5.63807591, 20.32530275, 14.25324163, 12.64785983,
       20.95671212, 24.91386093, -0.34555533, 13.62509538, 16.15367547,
       22.54865859, 25.00852603])
```

5.



Observations:

The linear regression model predicts housing prices with varying accuracy depending on the features. Residual plots help assess whether the model assumptions hold and highlight any non-linearity or heteroscedasticity.

Conclusion:

This experiment demonstrates the application of Linear Regression on the Boston Housing dataset. The model provides reasonable predictions, and visualization of residuals helps analyze model performance and possible improvements.

Viva Questions:

- What are residuals in regression analysis?
- Why is it important to visualize residuals vs predictions?
- How can overfitting be detected in regression models?
- What other metrics can be used to evaluate regression performance apart from MAE?